



Data Science Mentorship Programme

DatarFlow Founder Challenge 2026

The Niagara Succession Signal

Mapping small-business succession risk in Ontario's Niagara Region — and building the data products, forecasts, and remediation playbooks that a whole regional economy is currently missing.

Challenge Reference: DF-FC-2026-001

Issued: August 2026 · **Programme:** DatarFlow Cohort 2 (6-month internship)

Prize: ₦500,000 for the winning individual or team

Open: From Month 1 through the end of Month 5 of the internship (submissions close at the start of Month 6)

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1. Introduction

DatarFlow is a free, six-month data science mentorship programme for African professionals. Each cohort pairs structured learning with real, portfolio-grade work. This document sets out the DatarFlow Founder Challenge 2026: an optional, prize-backed research competition that runs alongside the internship and gives participants a chance to produce a genuine, publishable body of analytical work on a real-world problem.

The challenge asks participants — as individuals or in teams — to investigate a live and under-analysed question in the economy of the **Niagara Region of Ontario, Canada**: which small businesses are most at risk of closing not because they are failing, but because their owners are retiring with no successor and no buyer — and what should be done about it.

This is a data problem with a real audience. Niagara Region has, for the first time in 2026, begun collecting succession-planning data from local businesses. No one has yet turned that signal — combined with open national datasets — into a rigorous, forward-looking picture of where the risk sits, how many jobs are exposed, and what remediation looks like. That gap is the opportunity.

Why this challenge exists. Participants finish it with a published research report, a public dataset, infographics, and predictive models carrying their names — the kind of evidence that moves a career further than any certificate. DatarFlow gains a flagship analytical asset. And a real region gets analysis it currently lacks. Everyone's incentives point the same way.

A note on scope and geography. The core challenge is built around Niagara Region open data, all of which is accessible remotely from anywhere. Participants do **not** need to be in Canada. Teams that wish to may propose a parallel or comparative analysis of a Nigerian or other African regional economy using the same method — see Section 4.7. The mentorship team will confirm scope during onboarding.

2. Background: The Succession Signal

2.1 The national picture

Across Canada, a large share of small-business owners are approaching retirement, and most have no documented plan for what happens to the business when they leave. Independent research (see references, Section 9) has repeatedly found that:

- **Roughly three-quarters** of Canadian small-business owners hope to exit within a decade.
- The large majority have **no formal succession plan**; a substantial share have no plan of any kind.
- The single most-cited obstacle is **finding a suitable buyer or successor**.
- Well over a trillion dollars in business assets — and the jobs attached to them — sit behind these transitions.

When an owner retires and no buyer is found, the business simply closes. In the data, that closure is indistinguishable from a business that failed on its merits. The two call for opposite responses: a failing business may need demand support, while a healthy-but-orphaned business needs a buyer matched to it, ideally twelve to eighteen months ahead of the owner's exit. Nobody is currently drawing that distinction at the regional level.

2.2 Why Niagara, and why now

Niagara Region publishes an unusually rich local signal. Its annual Niagara Employment Inventory (NEI) sends teams to physically visit essentially every public-facing business in the region each summer. Recent cycles have covered well over thirteen thousand businesses at response rates above ninety percent, recording the jobs attached to each. Crucially, the 2026 inventory has, for the first time, added questions on succession planning.

That means two things. First, the region itself has signalled that succession is a live concern it wants data on. Second, the raw signal is arriving now and **has not yet been analysed** — which is precisely where this challenge sits. Niagara's business base is also concentrated in exactly the owner-operated sectors most exposed to succession risk: retail trade, accommodation and food services, and the cluster of "other services" (repair and maintenance, personal care, laundry, funeral services). These are consistently the region's largest sectors by business count and also among its highest closure counts.

The core question participants are answering:

"Which parts of the Niagara small-business economy are most exposed to owner-succession risk over the next 5–10 years; how many businesses and jobs does that represent; where is the risk concentrated by sector and municipality; and what specific, evidence-based actions should government and the private sector take to prevent avoidable closures?"

2.3 The hard line participants must respect

This challenge produces published regional analysis, not advice to any specific, named business owner.

Analysing aggregate open data and forecasting sector-level risk is ordinary, low-risk economic research. Advising an identified owner on how to sell their particular company, or brokering a specific sale, crosses into regulated business-brokerage and financial/tax-advisory territory. Participants must stay on the analytical side of that line at all times. Where a deliverable sounds transactional (for example, the "acquisition playbook" in Section 4), it must be written as a **generic framework**, never as a plan for a real, identifiable business.

3. Objectives

A strong submission will:

- Quantify and map owner-succession risk across the Niagara small-business economy, disaggregated by sector (NAICS) and by municipality (Niagara's twelve local-area municipalities).
- Estimate the number of businesses — and the number of jobs — attached to owners statistically likely to exit within the next 5–10 years.
- Distinguish, as far as the data allows, **succession-driven closure risk** from ordinary business-failure risk.
- Build a predictive/forecasting model that projects business closures or at-risk counts forward under stated assumptions, with honest treatment of uncertainty.
- Translate the findings into decision-ready outputs: infographics, an open dataset, and remediation recommendations aimed separately at government and at the private sector.
- Produce a generic, transferable acquisition/succession playbook that a prospective buyer or a policymaker could adapt — framed as a framework, not as advice for any specific business.
- Do all of the above to a standard, and with documentation, that would let an outside analyst reproduce the work.

4. Scope of Work & Required Assets

Every submission must include the core deliverables in 4.1–4.8. The "nice-to-have" extensions in 4.9 are optional and count toward the innovation score. Participants may scope sensibly given time and data — a well-justified narrower analysis beats an overreaching thin one.

4.1 Aggregated data layer

A cleaned, documented, aggregated dataset joining the relevant open sources (Section 8) at the sector and municipal level: business counts, business dynamics (entrants/exits), employment, sector composition, and any owner-age or succession proxies available. This is the analytical foundation every other deliverable draws on.

4.2 Research report

The central written deliverable: a structured research report presenting method, findings, and recommendations. It should stand on its own to an economic-development audience and follow the reproducibility and statistical-reporting standards in Section 6.

4.3 Infographics & visual assets

A set of clear, standalone infographics and charts communicating the headline findings to a non-technical audience — the risk map, the exposed-jobs figure, the sector and municipal breakdowns. These must be legible on their own, because they are what stakeholders actually circulate.

4.4 Graphs & exploratory visuals

The analytical visualisations behind the findings: distributions, trends, correlations, model diagnostics. Where the report makes a claim, a graph should support it.

4.5 Predictive model for forecasting business closures

A documented model that forecasts business closures or at-risk business counts (by sector and/or municipality) forward over a stated horizon. Participants must state assumptions, quantify uncertainty, validate honestly (hold-out or cross-validation, with baseline comparison), and be explicit about what the model can and cannot support. A transparent, well-validated simple model outscores an opaque complex one.

4.6 Business acquisition / succession playbook (generic)

A transferable playbook setting out how an at-risk but healthy business could be positioned for acquisition or orderly succession — the signals a buyer or matchmaker should look for, the sequence of a well-run transition, the common failure points. Written as a **generic framework informed by the data**, explicitly not as advice for any identifiable business (see 2.3).

4.7 Local-economy impact analysis

An analysis of what the modelled succession risk means for the Niagara economy in aggregate: jobs exposed, sectoral concentration, municipal vulnerability, knock-on effects, and which segments most repay early intervention.

4.8 Government & private-sector remediation report

A concise, decision-ready briefing translating findings into recommended actions, addressed separately to (a) government / economic-development bodies and (b) private-sector actors (buyers, lenders, brokers, chambers). Recommendations must trace to specific findings in the analysis.

4.9 Nice-to-have extensions (optional, scored for innovation)

- An interactive dashboard or web tool exposing the risk map and letting a user filter by sector and municipality.
- A succession-readiness index or scoring method usable year over year.
- A buyer–business matching approach (methodological, using aggregate/synthetic data — not real owner records).
- A comparative case study applying the same method to a **Nigerian or other African regional economy**, drawing out transferable lessons.
- A policy-brief version of the remediation report, formatted for circulation to decision-makers.
- A short reproducibility/replication package (code, environment, data-refresh instructions).

5. Eligibility & Team Formation

- Open to all currently enrolled DatarFlow Cohort 2 interns.
- **Individuals or teams** may enter. Teams may be self-formed and should be no larger than four members (the mentorship team may adjust this cap at onboarding).
- A participant may join only one submission. Team membership is fixed at the point of registration; changes require mentor approval.
- Participants register their intent to compete (solo or as a named team) at the start of the challenge window so mentors can track and support entries.
- Standard DatarFlow conduct, academic-honesty, and data-ethics rules apply. Plagiarism, fabricated results, or misuse of data is disqualifying.

6. Research, Data & Ethics Standards

These standards are drawn from DatarFlow's existing preprint-readiness practice and are graded directly (Section 7).

6.1 Reproducibility

Another analyst should be able to re-run the work. Submit code and clear instructions; fix random seeds; document every data source, version, and transformation; avoid manual, unrecorded steps.

6.2 Statistical reporting

Report uncertainty, not just point estimates. State assumptions and limitations plainly. Distinguish correlation from causation. For any model, give baseline comparison and honest validation. Do not over-claim what the data can support.

6.3 Data ethics & licensing

- Use only open or properly licensed data. Respect each source's terms and attribute every source.

- Work at aggregate level. Do not attempt to identify, profile, or single out any individual business owner or named business.
- Where data is thin, say so; do not fabricate or silently impute without disclosure.

6.4 The regulatory line

Re-read Section 2.3. Analysis and generic frameworks only — never transaction advice for a specific, identifiable business.

7. How Submissions Are Judged

A panel drawn from the DatarFlow mentorship team (including the Programme Director and Technical Lead) scores each submission against the weighted criteria below. The panel may award the prize to a single entry, split it between joint winners at its discretion, or — if no entry meets a minimum quality bar — decline to award it and instead recognise the strongest entries.

Criterion	What the panel looks for	Weight
Analytical rigour & method	Sound approach; correct handling of the data; defensible modelling choices	25%
Reproducibility & statistical reporting	Could an outsider re-run it? Honest uncertainty, assumptions, validation (Section 6)	20%
Completeness of required assets	Are the core deliverables in 4.1–4.8 all present and coherent?	15%
Insight & real-world usefulness	Do the findings and remediation actually help a decision-maker?	15%
Communication & visual quality	Clarity of report, infographics, and visuals for a non-technical audience	15%
Innovation (incl. 4.9 extensions)	Original, well-executed ideas beyond the core scope	10%

Tie-breaker. If two entries score within a hair of each other, the panel favours the one that is more reproducible and more immediately useful to a real Niagara stakeholder.

8. Where to Read & Where to Get the Data

Everything below is publicly and remotely accessible. Participants are not limited to these sources, but a strong entry will draw on the core datasets in 8.1. Always check each source's licence and cite it.

8.1 Core open datasets (the spine of the analysis)

Source	What it gives you	Where
Canadian Business Counts	Business location counts by NAICS industry and by employment-size band, down to census subdivision / CMA level — your base map of how many businesses of each type exist where. Updated twice yearly.	open.canada.ca — search "Canadian Business Counts"; StatsCan tables 33-10-0096-01 (with

Source	What it gives you	Where
		employees) & 33-10-0097-01 (CMA/CSD level)
Business Dynamics: Entrants, Incumbents & Exits	Counts of business entries and EXITS by industry and province from the Longitudinal Employment Analysis Program (LEAP) — the closest open proxy for closures to calibrate a forecasting model. Updated ~annually.	StatsCan table 33-10-0087-01 (open.canada.ca — "Business Dynamics measures by industry")
Niagara Employment Inventory (NEI)	Near-census of Niagara businesses with attached job counts, sector mix, closures/vacancies, top business concerns, and — from 2026 — succession-planning questions. The local signal at the heart of this challenge.	niagaracanada.com (Niagara Economic Development) → Data → Reports; some subsets are partner-only — see 8.4
Statistics Canada — StatCan / census	Owner/operator demographics, age structure of the self-employed, labour-force and industry data, and geography lookups for Niagara's municipalities.	www150.statcan.gc.ca (data tables & census profiles); www12.statcan.gc.ca for census-profile geographies
CFIB succession research	National baseline rates for succession-planning gaps, exit intentions, and obstacles — use to sanity-check and contextualise your Niagara estimates.	cfib-fcei.ca — search "succession"

8.2 Regional & municipal context

- **Niagara Economic Development** (niagaracanada.com) — economic updates, sector profiles, quick facts, and the NEI reports.
- **Niagara Region open data** (niagararegion.ca / niagaraopendata.ca) — municipal boundaries, geography, and regional statistics.
- **Niagara's twelve local-area municipalities** — St. Catharines, Niagara Falls, Welland, Fort Erie, Grimsby, Lincoln, Pelham, Port Colborne, Thorold, Wainfleet, West Lincoln, and Niagara-on-the-Lake — for municipal-level breakdowns.
- **Brock University & Niagara College** — regional economic research and reports; both receive NEI subsets and publish local analysis.

8.3 Background reading on the succession wave

- CFIB succession-readiness reporting and media coverage (Wealth Professional; The Logic).
- MNP LLP Succession Readiness Report and commentary (Talent Canada; On-Site Magazine).
- RBC Wealth Management and bank/accounting-firm guides on Canadian business succession.
- BizBuySell / IBBA Market Pulse and Canadian business-broker statistics for deal-volume and valuation context.
- Note the 2026 change to the Canadian capital-gains inclusion rate, which is reported to be compressing exit timelines — verify the current status before relying on it.

8.4 A note on the NEI succession data

Some NEI output is public; some is circulated only to municipal economic-development offices, the chambers, Brock, and Niagara College. If the raw 2026 succession responses are not public in time, participants should

model succession risk from open owner-age and business-dynamics proxies and treat the NEI as corroborating context. The mentorship team will share whatever NEI material it can obtain and will flag what is and isn't available at onboarding.

8.5 Tools

- Python (pandas, scikit-learn, statsmodels) or R for analysis and modelling; Jupyter/Quarto for reproducible notebooks.
- Any open charting/visualisation stack for infographics; keep source files so charts can be regenerated.
- Git for version control and to package the reproducibility bundle.

9. Timeline & Submission

The challenge runs from the start of the internship to the start of Month 6 — five months of working window, deliberately ending before the final month so the last weeks stay clear for programme wrap-up.

Phase	When	What happens
Launch & registration	Month 1	Challenge briefed; participants register solo/team intent; onboarding on scope, data access, and standards
Proposal checkpoint	End of Month 1	Each entry submits a one-page plan: question focus, data, intended assets, team roles
Build & mentor check-ins	Months 2–4	Core analysis and asset build; periodic mentor reviews mirroring the preprint-readiness process
Draft review	Mid Month 4	Optional draft submission for structured feedback (one round)
Final submission	Start of Month 6	Full submission package due (see 9.1). No late entries.
Judging & showcase	Month 6	Panel scores; finalists present; winner announced and prize awarded

9.1 What to submit

- A technical write-up / research report (4.2).
- The aggregated dataset (4.1) and reproducibility/code bundle.
- Infographics and visuals (4.3, 4.4).
- The forecasting model with documentation (4.5).
- The generic acquisition/succession playbook (4.6).
- The local-economy impact analysis (4.7) and the government & private-sector remediation report (4.8).
- Any 4.9 extensions attempted.
- A short declaration confirming original work, data sources used, and adherence to Section 6.

9.2 Format & how to submit

- Report and briefings as PDF; dataset in an open format (CSV/Parquet); code in a Git repository or zipped bundle with a README.
- Submit through the channel named at launch (programme portal or designated address). Confirm receipt.

10. Prize & Recognition

£500,000 awarded to the winning individual or team.

For a team, the prize is paid to the team to divide as its members agreed at registration; the mentorship team will ask each team to record an intended split up front to avoid disputes.

Beyond the cash:

- Winning and shortlisted work is published under DatarFlow as a flagship research asset, with participants credited as authors.
- Finalists present to the programme and invited stakeholders.
- Strong entries become portfolio centrepieces — a published regional analysis with named authorship, which is materially more valuable to a participant's career than a certificate.
- The panel may recognise standout individual assets (best model, best infographic set, best policy brief) even where they are not the overall winner.

11. Ownership, Publication & Confidentiality

- Participants retain authorship credit for their work at all times.
- By entering, participants grant DatarFlow a licence to publish, adapt, and disseminate submitted work as part of programme outputs and regional dissemination, with attribution.
- Winning and shortlisted work will be released publicly (report, anonymised/aggregated dataset, visuals) as an open contribution, consistent with the programme's public-good aim.
- All work must rely on open or properly licensed data; any partner-shared material (e.g. non-public NEI subsets) must be handled per the terms given at onboarding and never re-published beyond what those terms allow.
- No submission may include confidential, scraped-without-permission, or personally identifying business-owner information.

12. Questions & Contact

Questions about scope, data access, team formation, or the standards should go to the DatarFlow mentorship team through the usual programme channels. The Programme Director (operations) and Technical Lead (methodology, data, and modelling) are the points of contact for this challenge. A live Q&A will be held during the Month-1 launch, and clarifications will be shared with all registered participants so everyone works from the same information.

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